

An Edge-Deployable Probabilistic Conflict Analysis Framework: Summary of Methodology and Techniques

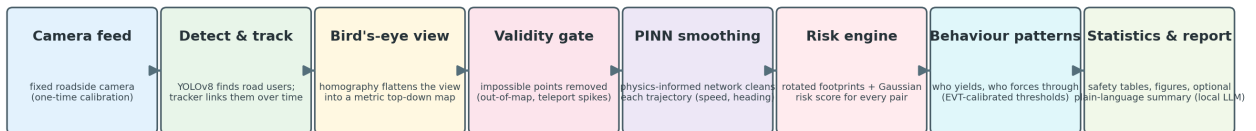
This document is a condensed technical summary of a camera-only conflict-analysis framework for automated road-safety monitoring. It describes the methodology and the techniques applied at each stage of the pipeline, defines every quantity the framework computes, and states how each one is calculated. Classical surrogate safety measures such as Time-to-Collision (TTC) and Post-Encroachment Time (PET) model road users as points or line-crossings moving at constant velocity and reduce an encounter to a binary safe/conflict verdict. The framework summarised here replaces each of those assumptions: constant-acceleration trajectory reconstruction by a physics-informed neural network in place of constant velocity; rotated two-dimensional polygon footprints with a Separating-Axis-Theorem collision sweep in place of point-mass agents; a bounded, continuous Criticality Index derived from a bivariate Gaussian occupancy field in place of a binary conflict flag; and joint treatment of all class-valid road-user pairs (pedestrian–vehicle, vehicle–vehicle, cyclist–vehicle) in place of single-interaction-type analysis. Behavioural interpretation is added through a role-neutral conflict-zone buffer and two derived negotiation descriptors, with classification thresholds calibrated by extreme value theory. Only methodology is presented; no data or results are reported.

Index Terms—Surrogate safety measures, probabilistic collision risk, physics-informed neural network, homography rectification, separating axis theorem, extreme value theory, edge computing.

1 Introduction

Automated safety monitoring at unsignalized, mixed-traffic junctions must run continuously, cover heterogeneous road users, and operate within the compute budget of a roadside unit (RSU). Classical surrogate safety measures were not designed under these constraints. They assume point-like agents moving at constant velocity, they collapse a continuously evolving encounter into a single scalar or a binary verdict, and their governing formulas are undefined for near-parallel headings or near-zero speeds — precisely the regime of hesitation, yielding, and contested right-of-way that dominates informal negotiation.

The framework summarised here is organised as a streaming, single-pass pipeline (Fig. 1) whose stages are individually cheap enough for edge deployment: detection and tracking; homography-based projection into a metric bird’s-eye view (BEV); a validity gate on raw projected detections; per-track physics-informed smoothing; a pairwise risk engine combining polygon geometry with a Gaussian occupancy field; a behavioural classification layer with statistically calibrated thresholds; and a statistical/reporting layer. Each stage and its mathematics are described below, and Section 4 tabulates every computed quantity.



Everything runs on a small computer at the roadside — only compact risk summaries leave the device, never raw video.

Figure 1: Pipeline organisation. Each stage is chosen for per-frame, single-pass operation so the chain can run on edge-class hardware, with only compact per-encounter risk summaries transmitted upstream.

2 System Architecture

2.1 Perception: Detection and Tracking

A single-pass object detector (YOLOv8 class) operating at a fixed confidence threshold classifies each frame into eight road-user classes: pedestrian, cyclist, motorcycle, auto-rickshaw, car, light commercial vehicle, bus, and truck. A multi-object tracker associates detections across frames so each agent retains a stable identity while visible, and a polygon region-of-interest filter restricts processing to the roadway. Detector class labels are remapped onto the eight-class taxonomy through an explicit mapping layer, so that footprint geometry downstream is always tied to a physically meaningful vehicle category.

2.2 Metric Rectification: Homography and Scale

An image-plane observation carries no metric information: perspective compresses distance non-uniformly. A homography H — a 3×3 projective transform estimated once per camera installation — maps image pixels onto a flat, top-down canvas in which relative distances are preserved:

$$\begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = H \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}, \quad (x_{\text{px}}, y_{\text{px}}) = \left(\frac{u'}{w'}, \frac{v'}{w'} \right). \quad (1)$$

The standard estimation route uses four coplanar correspondences of known geometry. When no rectangular ground feature of known dimensions is available — common for strongly oblique installations — rectification can instead be obtained in two stages. First, a projective correction is derived from two vanishing points estimated from parallel scene structure, which removes perspective convergence but leaves a residual affine distortion. Second, that residual is removed using a circular scene feature of known radius: a circle viewed obliquely projects to an ellipse, and the transform that maps the fitted ellipse back to a circle is exactly the metric correction required. The composed transform H is applied per frame at negligible marginal cost.

Metric scale follows from any one known real-world length. With a circular feature of true radius R_m whose fitted radius in the rectified canvas is R_{px} , the scale is

$$s = \frac{R_{\text{px}}}{R_m} \quad [\text{px/m}], \quad (x, y) = \left(\frac{x_{\text{px}}}{s}, \frac{y_{\text{px}}}{s} \right), \quad (2)$$

so that every downstream quantity — separations, footprints, speeds, buffers — is expressed in metres and seconds. A misestimated s rescales every distance in the pipeline by the same factor, so this constant is the single most safety-critical calibration parameter.

2.3 Track Validity Gating

Projective rectification amplifies error away from the calibration anchor, producing two characteristic artifacts: detections near the image horizon that project to implausible distances, and single-frame outliers where pixel-level jitter in a magnified region implies impossible displacement. Both are removed before any physical quantity is estimated, by two rules applied to raw projected detections.

Spatial validity. Points falling outside the bounded region for which the homography is metrically reliable are discarded.

Kinematic spike gate. For a track sampled at times t_i with positions $\mathbf{p}_i$, define the implied speed to each neighbour, $v_i^- = \|\mathbf{p}_i - \mathbf{p}_{i-1}\|/(t_i - t_{i-1})$ and $v_i^+ = \|\mathbf{p}_{i+1} - \mathbf{p}_i\|/(t_{i+1} - t_i)$. A sample is rejected when

$$v_i^- > V_{\text{gate}} \quad \text{and} \quad v_i^+ > V_{\text{gate}}, \quad (3)$$

with V_{gate} a physical speed bound for the site. Requiring *both* neighbours to exceed the bound isolates single-sample excursions while preserving genuinely fast agents, whose consecutive steps are individually plausible.

2.4 Deployment Model and Reporting Layer

Every stage is constant-cost per frame or per agent pair with no iterative search: detection and projection are per-frame operations, smoothing is a small per-track fit, and the risk engine is closed-form, so only compact

per-encounter summaries need leave the edge node. An optional final layer renders the structured per-encounter table into operator-readable natural-language text using a locally hosted language model; it is advisory, the structured output remaining authoritative.

3 Methodology

3.1 Constant-Acceleration Trajectory Reconstruction (PINN)

Raw positions remain noisy after gating, and differentiating noise directly corrupts the velocity, acceleration, and heading estimates on which every later metric depends. Each track is therefore smoothed independently by a small *physics-informed neural network* (PINN) — a network trained not only to fit data but to respect a governing physical constraint. A network $f_\theta : t \mapsto (x, y)$ (two hidden layers of tanh units) is fit per track by minimising

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{phys}}, \quad \mathcal{L}_{\text{data}} = \frac{1}{N} \sum_{i=1}^N \|f_\theta(t_i) - \mathbf{p}_i\|^2, \quad (4)$$

$$\mathcal{L}_{\text{phys}} = \text{mean} \left(\text{ReLU}(\|\mathbf{a}(t)\| - a_{\max}) \right), \quad \mathbf{a}(t) = \frac{d^2 f_\theta}{dt^2}, \quad (5)$$

where $\mathbf{a}(t)$ is obtained by twice finite-differencing the network’s own output and $a_{\max}$ is a plausible acceleration bound for road users. The hinge form penalises only violations, so ordinary motion is unpenalised while physically impossible jerk is expensive. Time and position are normalised per track before fitting, and true inter-sample intervals are used in the differences, so irregular sampling is handled correctly. Velocity, acceleration, and heading are then read from the smoothed sequence by finite differencing with light temporal averaging:

$$\mathbf{v}_i = \frac{\mathbf{p}_i^{\text{sm}} - \mathbf{p}_{i-1}^{\text{sm}}}{\Delta t_i}, \quad \theta_i = \arctan 2(v_{y,i}, v_{x,i}). \quad (6)$$

The result is a trajectory consistent with constant-acceleration motion, replacing the constant-velocity assumption of classical measures while preserving genuine braking and acceleration events.

3.2 Rotated Polygon Footprints and SAT Separation

Each agent is represented not as a point but as a rectangle of class-specific width w and length l (Table 1), centred at its smoothed position and rotated to its heading:

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad \mathbf{c}_k = R(\theta) \mathbf{c}_k^{\text{local}} + (x, y), \quad (7)$$

for the four local corners $\mathbf{c}_k^{\text{local}} = (\pm l/2, \pm w/2)$.

Separation between two convex polygons is evaluated with the *Separating Axis Theorem* (SAT): two convex shapes are disjoint if and only if there exists an axis on which their projections do not overlap, and for rectangles it suffices to test the edge normals of both. Projecting each polygon onto a candidate unit axis $\mathbf{n}$ gives intervals $[\min_k \mathbf{c}_k^A \cdot \mathbf{n}, \max_k \mathbf{c}_k^A \cdot \mathbf{n}]$ and likewise for B ; the signed gap on that axis is

$$g(\mathbf{n}) = \max \left(\min_k \mathbf{c}_k^B \cdot \mathbf{n} - \max_k \mathbf{c}_k^A \cdot \mathbf{n}, \min_k \mathbf{c}_k^A \cdot \mathbf{n} - \max_k \mathbf{c}_k^B \cdot \mathbf{n} \right). \quad (8)$$

The true minimum separation is the *maximum* gap over all candidate axes, $d_{\text{sep}} = \max_{\mathbf{n}} g(\mathbf{n})$ when any gap is positive, and zero (intersection) otherwise. This distinction matters: the first axis on which a gap is found is not generally the axis of minimum separation, so an early return understates clearance. The polygon representation captures what a centre-to-centre distance cannot — an elongated vehicle passing broadside presents a very different physical clearance than the same centroid distance end-on.

3.3 Constant-Acceleration Collision Sweep

Predictive collision timing is obtained by forward-simulating both agents under their current kinematics,

$$x(t) = x_0 + v_x t + \frac{1}{2} a_x t^2, \quad y(t) = y_0 + v_y t + \frac{1}{2} a_y t^2, \quad (9)$$

Table 1: Class-specific footprint dimensions. The same values are used by the polygon geometry, the Gaussian occupancy field, and the conflict-zone buffer, so all components reason about identical physical agent sizes.

Class	Width (m)	Length (m)
Pedestrian	0.50	0.50
Cyclist	0.60	1.80
Motorcycle	0.75	2.00
Auto-rickshaw	1.30	2.65
Car	1.50	3.50
Light commercial vehicle	1.80	4.50
Bus	2.00	5.00
Truck	2.00	5.00

and applying the SAT test of Eq. (8) to the forward-projected footprints at fixed time steps up to a bounded horizon. The *polygon time-to-collision* TTC_{poly} is the first step at which the predicted footprints intersect while the agents are still converging (centre distance decreasing). If no intersection is predicted within the horizon the value is deliberately undefined: this encodes “no predicted collision”, which is a meaningful outcome rather than a failure.

3.4 Bivariate Gaussian Occupancy and the Continuous Criticality Index

To obtain a risk measure that is defined under *every* geometry, each agent’s spatial presence is modelled as a bivariate Gaussian centred on its position, with standard deviations set from half its class-specific dimensions (subject to a floor). For two agents with combined variances $\sigma_x^2 = \sigma_{x1}^2 + \sigma_{x2}^2$ and $\sigma_y^2 = \sigma_{y1}^2 + \sigma_{y2}^2$ and separation $(\Delta x, \Delta y)$, the *Continuous Criticality Index* is the joint density at the observed separation normalised by its value at co-location:

$$CCI = \exp\left(-\frac{1}{2} \left[\frac{\Delta x^2}{\sigma_x^2} + \frac{\Delta y^2}{\sigma_y^2} \right]\right) \in (0, 1]. \quad (10)$$

CCI equals 1 at exact co-location and decays smoothly with separation, scaled by combined physical size, so larger vehicles acquire a proportionally wider high-criticality region. Being a closed-form exponential of instantaneous position, it has no geometric degeneracy: it remains defined for parallel headings, zero speed, and diverging paths, where boundary-crossing formulations return no value. The per-encounter statistic reported is max CCI, the peak reached over the shared observation window.

3.5 Conflict Zone, Exit Buffer, and Negotiation Descriptors

Behavioural interpretation requires knowing not merely how close two agents came, but how the right-of-way was resolved. A conflict point is solved as the intersection of the two agents’ heading rays. For each agent the second-order kinematic time to travel a distance d from rest-relative speed v_0 under acceleration a follows from inverting $d = v_0 t + \frac{1}{2} a t^2$:

$$t = \frac{-v_0 + \sqrt{v_0^2 + 2ad}}{a} \quad (a \neq 0), \quad t = \frac{d}{v_0} \quad (a \rightarrow 0). \quad (11)$$

Applying Eq. (11) to the near and far edges of the shared conflict zone — using each agent’s true footprint extent along its path rather than a point — yields arrival and clearance times, whose differences define the *Exit Buffer* (PEB): the signed time margin separating the two agents’ occupation of the conflict zone. The sign indicates which agent is currently ahead in the negotiation; the outcome of the encounter is read from the sign structure at the last frame at which both boundary values are defined, giving three cases: agent A clears first, agent B clears first, or the zone remains unresolved within the observation window.

Two descriptors summarise the temporal evolution of PEB. Let $\{(t_j, \text{PEB}_j)\}$ be the valid samples of the mean buffer signal.

Negotiation Momentum (NM) is the typical rate of change of the buffer, computed as the median of

per-sample rates against true elapsed time:

$$\text{NM} = \text{median}_j \left\{ \frac{\text{PEB}_j - \text{PEB}_{j-1}}{t_j - t_{j-1}} \right\}. \quad (12)$$

The median is used in place of a mean because a small number of extreme per-sample rates near geometric degeneracies would otherwise dominate short windows; using true elapsed time prevents gaps in valid samples from inflating the rate.

Sustained Dominance Duration (SDD) is the elapsed time of the most recent unbroken run of constant buffer sign:

$$\text{SDD} = t_{\text{last}} - t_{j^*}, \quad j^* = \min\{j : \text{sgn}(\text{PEB}_k) = \text{sgn}(\text{PEB}_{\text{last}}) \forall k \geq j\}. \quad (13)$$

Together, NM answers “is the advantage building toward one agent or oscillating?” and SDD answers “how long has the eventual winner’s advantage been stable?”

Post-Encroachment Time (PET) is measured from observed rather than predicted geometry: if the two footprints ever genuinely overlap during the window, a direct encroachment is recorded; otherwise PET is derived from the trajectory around the instant of closest approach, as the residual time margin about that critical instant.

All of these quantities are computed for every class-valid agent pair (pedestrian–vehicle, vehicle–vehicle, cyclist–vehicle; pedestrian–pedestrian excluded) using an agent-role-neutral “agent A / agent B” formulation, so one implementation covers every interaction type. Pairs are admitted for analysis only when they are simultaneously visible, within a proximity radius, and share a minimum duration of valid observation.

3.6 Threshold Calibration by Extreme Value Theory

Classifying dominance as “stable” or “marginal” requires a threshold on SDD. Rather than fixing one by hand, it is calibrated from the data by *peaks-over-threshold* extreme value theory, which models exceedances above a high threshold u by the Generalized Pareto Distribution

$$G_{\xi, \sigma}(y) = 1 - \left(1 + \frac{\xi y}{\sigma}\right)^{-1/\xi}, \quad y = X - u > 0, \quad (14)$$

with shape ξ and scale σ . A mean-residual-life scan evaluates the mean exceedance across candidate thresholds, and a GPD fit is performed at each candidate; the selected threshold is the one beyond which the fitted parameters stabilise, which is the standard diagnostic for the regime in which the asymptotic tail model holds. Thresholds are calibrated separately per pair type when the sample supports it, with a fixed fallback otherwise. Complementarily, NM is classified against a neutral band scaled to the observed dispersion of that pair type, $\pm \kappa \sigma_{\text{NM}}$, so “momentum toward A”, “neutral”, and “momentum toward B” are defined relative to the data rather than by an arbitrary constant.

3.7 Behavioural Pattern Classification

The triple (outcome, NM sign relative to the neutral band, SDD relative to its calibrated threshold) maps each encounter onto an interpretable label (Table 2). Pedestrian–vehicle pairs use pedestrian-semantic labels; vehicle–vehicle and cyclist–vehicle pairs use role-neutral equivalents. Encounters whose buffer never resolves within the observation window are labelled explicitly as unknown rather than being forced into a pattern.

3.8 Statistical Analysis

Pattern-level safety differences are assessed with a linear mixed-effects model fitted to the pooled safety indicators. Writing y for a stacked indicator value (PEB, PET, or TTC), the model is

$$y_{ij} = \beta_0 + \mathbf{x}_{ij}^\top \boldsymbol{\beta} + u_j + \varepsilon_{ij}, \quad u_j \sim \mathcal{N}(0, \tau^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2), \quad (15)$$

with fixed effects $\mathbf{x}$ encoding the behavioural pattern and the indicator type (dummy-coded against reference levels) and a random intercept u_j for road-user class, which absorbs class-level heterogeneity so that pattern effects are not confounded by vehicle type. Extreme values of the buffer-type indicators are clipped before fitting to bound the influence of degenerate near-stationary geometry. The random-intercept variance τ^2 is itself informative: a negligible value indicates that once the negotiation pattern is known, the vehicle class adds little further explanatory power.

Table 2: Behavioural pattern taxonomy. Each label is determined by the encounter outcome combined with the momentum band and the EVT-calibrated dominance threshold.

Pattern	Momentum	Dominance	Interpretation
Proactive Yielding	toward A	stable	one party gives way early and consistently
Marginal Yield	toward A	marginal	yields, but advantage never stabilises
Gap-Driven Crossing	toward B	stable	crossing party exploits a sustained opening
Late Pedestrian Surge	toward B	marginal	crossing initiated late, little margin
Insufficient Yield	toward A	stable	nominal yield leaves inadequate buffer
Pedestrian Hesitation	toward A	marginal	stop-start behaviour in the conflict zone
Assertive Non-Yielding	toward B	stable	one party presses through without yielding
Contested Non-Yielding	toward B	marginal	neither party concedes; margin unstable
Agent A/B Priority (Stable / Marginal)	toward A/B	stable / marginal	role-neutral equivalents for vehicle- and cyclist-vehicle pairs
Gap-Forced Merge	toward B	stable	merging party forces into a closing gap
Contested Right-of-Way	toward B	marginal	priority disputed throughout
Unclear Advantage	neutral	stable	resolved without decisive dominance
Chaotic Negotiation	neutral	marginal	advantage oscillates; no stable resolution
Unknown	—	—	buffer never resolves within the observed window

4 Summary of Computed Quantities

5 Design Rationale and Limitations

Three properties motivate the design. *Coverage*: the criticality index is a closed-form exponential and therefore always defined, whereas boundary-crossing formulations legitimately return no value under near-parallel or low-speed geometry — an operational problem for a monitor that must distinguish “computed safe” from “could not compute”. *Fidelity*: rotated footprints recover clearance and critical timing that centroid-distance proxies structurally cannot represent for extended vehicles. *Interpretability*: the buffer-derived descriptors turn a proximity measurement into a statement about how right-of-way was negotiated, with statistically rather than manually chosen thresholds.

The limitations follow from the same structure. Behavioural labels inherit identity errors from tracking, since an encounter is only as meaningful as the two trajectories defining it; metric accuracy degrades with distance from the calibration anchor under oblique viewing; heading, and hence conflict-point geometry, is least reliable for nearly stationary agents, whose direction derives from a noise-dominated velocity; the smoother is a per-track batch fit rather than a recursive filter, so detection gaps are interpolated by construction and the residual risk is that an identity switch yields a smooth but wrong trajectory rather than a failed one; and admitting pairs by proximity alone can include agents near one another without genuinely interacting, so requiring evidence of engagement (unobstructed line of sight, converging paths) is the natural refinement.

6 Conclusion

The framework summarised here replaces the point-mass, constant-velocity, binary-verdict assumptions of classical surrogate safety analysis with constant-acceleration reconstruction by a physics-informed network, exact rotated-polygon collision geometry, a bounded continuous criticality index, and a role-neutral behavioural layer whose thresholds are calibrated by extreme value theory — all within a per-frame, closed-form cost profile suitable for edge deployment. Each technique and every quantity it produces is defined above; no data or results are reported in this summary.

7 Novelty Relative to the Original DCAF Formulation

The behavioural layer of this work builds on the Dynamic Conflict Analysis Framework (DCAF) of Muduli and Ghosh [1], which established the conceptual triad retained here: a continuously defined conflict-zone exit buffer, its directional rate of change (Negotiation Momentum), and the persistence of the resulting advantage

Table 3: Every quantity the framework computes, with its definition and method of calculation.

Quantity	Meaning	How it is calculated
H, s	projective transform to the metric plane; pixel-per-metre scale	two-stage rectification (vanishing points, then ellipse-to-circle); s from a feature of known real size, Eqs. (1)–(2)
Smoothed track	position, velocity, acceleration, heading per agent	per-track PINN fit under an acceleration-bound penalty, then finite differencing, Eqs. (4)–(6)
Footprint	rotated rectangle of true class dimensions	$R(\theta)$ applied to local corners, Eq. (7), Table 1
d_{sep}	true minimum clearance between two agents	SAT: maximum projected gap over all candidate axes, Eq. (8)
TTC_{poly}	predicted time until footprints intersect	constant-acceleration forward sweep with SAT at each step, Eq. (9); undefined if no collision predicted
TTC	classical kinematic boundary-crossing time	second-order arrival times at the conflict-zone boundaries, Eq. (11)
CCI	bounded continuous criticality in $(0, 1]$	normalised bivariate Gaussian occupancy overlap, Eq. (10); per encounter, the maximum over the window
PEB	signed time margin separating conflict-zone occupation	second-order arrival/clearance times at near and far zone edges using footprint extent, Eq. (11)
Outcome	which agent resolved the conflict first	sign structure of the buffer pair at the last fully valid frame; otherwise unresolved
NM	negotiation momentum	median of per-sample buffer rates against true elapsed time, Eq. (12)
SDD	sustained dominance duration	elapsed time of the latest constant-sign run of the buffer, Eq. (13)
PET	post-encroachment time	from observed footprint overlap, or the residual margin about closest approach
Pattern	behavioural label for the encounter	(outcome, NM band, SDD vs. EVT threshold) mapped through Table 2
u^*, ξ, σ	calibrated dominance threshold and GPD tail parameters	mean-residual-life scan plus GPD fits; threshold where parameters stabilise, Eq. (14)
β, τ^2	pattern-level effects and class-level variance	linear mixed-effects fit to pooled indicators, Eq. (15)

(Sustained Dominance Duration), together with extreme-value calibration of the dominance threshold and a mixed-effects assessment of pattern-level safety implications. That contribution is the behavioural foundation on which the present framework stands, and it is retained deliberately rather than re-derived.

What is new here is the physical and computational substrate beneath that behavioural layer, and the operational envelope it enables. Table 4 summarises the differences; the paragraphs below explain why each matters.

7.1 From constant velocity to constant acceleration

The original conflict-point construction projects each agent forward along its current heading at its current speed. Negotiation, however, is expressed almost entirely through *changes* of speed: a driver easing off, a pedestrian breaking stride, a rider accelerating into a gap. A constant-velocity projection is structurally blind to precisely the signal the behavioural layer is trying to read. Reconstructing each track with an acceleration-aware model, and carrying that acceleration into every arrival time (Eq. (11)) and every forward prediction (Eq. (9)), makes the buffer respond to yielding and surging as they happen rather than one sampling interval later.

7.2 From boundary geometry to full footprint geometry

Representing the pedestrian as a point and the vehicle through its front edge is a considerable advance over treating both as points, but it remains an asymmetric, partial description: it captures where the vehicle’s

Table 4: Methodological differences between the original DCAF formulation and the framework summarised in this document.

Aspect	Original DCAF formulation	This framework
Motion model	straight-line projection from instantaneous position and velocity (path assumed unchanged)	constant-acceleration trajectories reconstructed by a physics-informed network, Eqs. (4)–(6)
Agent geometry	pedestrian as a point; vehicle through its front-edge boundary corners	both agents as rotated rectangles of true class dimensions, Eq. (7)
Separation measure	distances to conflict-zone boundaries	exact convex separation by SAT, maximum gap over all axes, Eq. (8)
Collision prediction	kinematic boundary-crossing times	additional forward polygon sweep predicting <i>whether and when</i> footprints intersect, Eq. (9)
Risk representation	categorical zone state from buffer signs	additional bounded continuous index in $(0, 1]$, defined for every geometry, Eq. (10)
Interaction scope	pedestrian–vehicle pairs	all class-valid pairs (pedestrian–vehicle, vehicle–vehicle, cyclist–vehicle) via a role-neutral agent A/B formulation
Pattern taxonomy	ten pedestrian-semantic patterns	extended with role-neutral priority/merge labels and an explicit unresolved category (Table 2)
Trajectory source	commercial offline analytics platform, resampled to a uniform interval	self-contained detection, tracking and rectification at the native sampling rate, with the time base derived from the data
Rectification	correspondences on known ground geometry	two-vanishing-point plus ellipse-to-circle rectification for oblique views with no rectangular target, Sec. 2.2
Input validation	—	explicit two-rule validity gate on projected detections, Eq. (3)
Momentum estimator	mean rate of buffer change	median rate against true elapsed time between valid samples, Eq. (12)
Operational envelope	offline analysis of pre-extracted trajectories	per-frame, closed-form pipeline sized for edge deployment, with optional on-device natural-language reporting

front will be, not the space either body actually occupies. Modelling both agents as rotated rectangles and measuring their true convex separation (Eq. (8)) resolves clearances that a boundary construction cannot — most obviously for long vehicles encountered obliquely, where the governing gap is along the flank rather than the front. It also changes *when* an encounter is judged critical, because the instant of minimum footprint separation need not coincide with the instant of minimum centre-to-centre or front-edge distance.

7.3 From categorical states to an always-defined continuous index

In the original formulation, criticality is expressed through the sign structure of the buffer pair: opposite signs denote a contested zone state. This is behaviourally meaningful but categorical, and it inherits the degeneracies of the underlying kinematic solution — near-parallel headings or near-zero speeds leave the boundary times, and hence the state, undefined. The Gaussian criticality index of Eq. (10) is a closed-form exponential of instantaneous position scaled by physical size: it is bounded, graded, and defined for every configuration without exception. The two are complementary — the categorical state says *who is winning*, the continuous index says *how close it is* — and having both means the framework never has to report silence where a monitor needs a number.

7.4 From one interaction type to the full pair space

Restricting analysis to pedestrian–vehicle encounters excludes most of what occurs in heterogeneous traffic: two-wheelers filtering past each other, cyclists negotiating with light commercial vehicles, larger vehicles resolving priority among themselves. Recasting the buffer and its descriptors in role-neutral agent A/agent B terms allows one implementation to cover every class-valid pairing, with the pattern vocabulary extended

accordingly. The pedestrian-semantic labels are preserved where they apply, so comparability with the original taxonomy is not lost.

7.5 From offline study instrument to deployable monitor

The original evaluation consumes trajectories produced by a commercial analytics platform and resampled to a uniform interval. The framework here performs detection, tracking, rectification, validation, and smoothing itself, at the native sampling rate, with the time base inferred from the data — and every subsequent stage is constant-cost per frame or per agent pair with no iterative search. Two further additions serve the same goal: the two-stage rectification of Sec. 2.2, which makes metric analysis possible on oblique installations lacking a rectangular calibration target, and the validity gate of Eq. (3), which prevents projective and tracking artifacts from entering physical computation at all. The result is a framework that can run beside the camera rather than in a post-processing session, which is what turns a behavioural analysis method into a continuous safety-monitoring capability.

7.6 Summary

The original DCAF contributed the insight that right-of-way negotiation has measurable structure, and the indicators that expose it. This framework keeps that insight intact and rebuilds everything underneath it: the motion model, the geometry, the risk representation, the scope of interactions, the data path, and the deployment profile. The behavioural question is inherited; the physical and computational answer to it is new.

References

- [1] K. Muduli and I. Ghosh, “A dynamic conflict-analysis framework for behaviour-centric safety evaluation of pedestrian–vehicle interactions at unsignalized crosswalks,” *Accident Analysis and Prevention*, vol. 231, art. 108484, 2026.